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Gradient Extremum Seeking With Nonconstant Delays

GEORGE CARNEIRO SANTOS AND TIAGO ROUX OLIVEIRA[✉], (Senior Member, IEEE)

Department of Electronics and Telecommunication Engineering, State University of Rio de Janeiro (UERJ), Rio de Janeiro 20550-900, Brazil

Corresponding author: Tiago Roux Oliveira (tiagoroux@uerj.br)

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ABSTRACT This paper proposes a gradient extremum seeking method to address locally quadratic static maps in the presence of time-varying delays. Accommodating nonconstant delays has a strong impact in the predictor construction in terms of the associated transport partial differential equation (PDE) with variable convection speeds beyond the restrictions imposed on the delay regarding its arbitrary duration but bounded variation. A novel predictor design using perturbation-based estimates of the unknown Gradient and Hessian of the map must be introduced to handle this variable nature of the delays, which can arise both in the input and output channels of the nonlinear map to be optimized. Local exponential stability and convergence to a small neighborhood of the unknown extremum point are guaranteed. This technical result is assured by using backstepping transformation and averaging theory in infinite dimensions. Implementation aspects of the presented predictor for variable delays as well as the extension to state-dependent delays are also discussed. At last, we introduce the first results in the topic of extremum seeking control for cascades of transport PDEs that are interconnected through boundary conditions. Such a PDE-PDE cascade is useful to represent simultaneous time- and state-dependent delays. A simulation example illustrates the effectiveness of the proposed predictor-based extremum seeking approach for time-delay compensation.

INDEX TERMS Extremum seeking, time-varying delays, state-dependent delays, predictor feedback, partial differential equations, averaging in infinite dimensions, backstepping transformations.

I. INTRODUCTION

Extremum Seeking (ES) is a sort of adaptive control methodology which does not follow directly the standard paradigm of reference model, where the control objective of stabilizing the error between the reference trajectory and the output signal of the plant with known parametrization is studied [1]–[3]. Generally speaking, ES is a real-time optimization strategy which does not assume any explicit knowledge a priori of the static map and/or the dynamic plant to be optimized, but considers at least that such a plant has an optimal point (minimum or maximum) [4]. By employing periodic dithers (sinusoidal perturbations), the estimation in the average sense of the Gradient and the Hessian of the nonlinear map is attained so that the local stability proof for the overall closed-loop system and the convergence to a small neighborhood of the extremum can be assured.

In the last two decades, lots of theoretical advances and various applications have been found in the ES

literature [5]–[14]. Nevertheless, in the majority of the engineering fields, sensor and actuator signals are corrupted by delays [15]. Of course, depending on the specific application, such kind of delays may be: known/unknown, small/large, deterministic/stochastic or constant/variable [16], [17]. The latter type of classification is the main topic of interest in this paper.

Publication [18] has shown that a small amount of (constant) delay is enough to make the closed-loop feedback unstable when classic scalar gradient ES is solely considered. In this same reference, a solution to the delay problem was proposed by following two distinct decisions: (i) the inclusion of a predictor feedback in the ES loop by incorporating averaging-based estimates for the unknown Gradient's and Hessian's map, and (ii) the phase compensation of the additive perturbation signal.

Thereafter, extensions for Newton-based ES [19] with faster responses, optimization of the map's higher-derivatives [20], maximization of dynamic maps [21], expansions for multiparameter ES with output delays (or the same delay in the distinct input channels) [22] and multivariable ES under

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multiple and different input delays [23] were carried out. All theoretical foundation given in [18]–[23] could be conducted regarding that the attacked time-delays were *constant*.

The main goal of this paper is to generalize our previous results on prediction-feedback for ES in order to include a more general class of variable delays, not only time-varying delays but delays depending on the system states and time, simultaneously. Original results are pursued with respect to the existing literature as well as the improvement of the performance of the ES control systems under a quite wide class of variable delays. We introduce the fundamental steps to demonstrate local stability properties for nonlinear (locally quadratic) maps by means of averaging tools in infinite dimensions [24], [25], since the whole system is infinite dimensional due to the delays, and backstepping transformations [16]. A relevant debate considering the applicability of previous averaging results (oriented to fixed delays) is also derived, by identifying current limitations and possible showstoppers. Finally, numerical simulations for an academic example corroborate to the validation of the new ES algorithm.

Different from its earlier conference version [26], this paper includes the discussion about the numerical implementation of the proposed predictor for time-varying delays beyond carrying out additional numerical results. Moreover, we have corrected a critical error in the previous stability proof and introduce the extension of the gradient extremum seeking approach to the class of state-dependent delays. We also shed light in new opportunities for generalizing the results of extremum seeking to PDE-PDE cascades of transport type with constant and variable convection speeds, being an innovative approach for representing time- and state-dependent delays.

II. PRELIMINARIES

The 2–norm of a finite-dimensional (ODE) state vector $X(t)$ is denoted by single bars, $|X(t)|$. In contrast, norms of functions (of x) are denoted by double bars. By default, $\|\cdot\|$ denotes the spatial $L_2[0, D]$ norm, *i.e.*, $\|\cdot\| = \|\cdot\|_{L_2[0, D]}$. Since the PDE state variable $u(x, t)$ is a function of two arguments, we should emphasize that taking a norm in one of

the variables makes the norm a function of the other variable, as adopted in [16]. For example, the $L_2[0, D]$ norm of $u(x, t)$ in $x \in [0, D]$ is $\|u(t)\| = \left(\int_0^D u^2(x, t) dx\right)^{1/2}$.

Now, consider a generic nonlinear system $\dot{x} = f(t, x, \epsilon)$, where $x \in \mathbb{R}^n$, $f(t, x, \epsilon)$ is periodic in t with period T , *i.e.*, $f(t + T, x, \epsilon) = f(t, x, \epsilon)$. Thus, for $\epsilon > 0$ sufficiently small, we can obtain its average model given by $\dot{x}_{av} = f_{av}(x_{av})$, with $f_{av}(x_{av}) = 1/T \int_0^T f(\tau, x_{av}, 0) d\tau$, where $x_{av}(t)$ denotes the average version of the state $x(t)$ [27].

As defined in [27], a vector function $f(t, \epsilon) \in \mathbb{R}^n$ is said to be of order $\mathcal{O}(\epsilon)$ over an interval $[t_1, t_2]$ if there exist positive constants k and ϵ^* such that $|f(t, \epsilon)| \leq k\epsilon$, $\forall \epsilon \in [0, \epsilon^*]$ and $\forall t \in [t_1, t_2]$. In most cases we provide no precise estimates for the constants k and ϵ^* , and we use $\mathcal{O}(\epsilon)$ to be interpreted as an order of magnitude relation for sufficiently small ϵ .

The term “ s ” stands either for the Laplace variable or the differential operator “ d/dt ”, according to the context. For a transfer function $H_0(s)$ with a generic input u , pure convolution $h_0(t) * u(t)$, with $h_0(t)$ being the impulse response of $H_0(s)$, is also denoted by $H_0(s)u$, as done in [28].

III. PROBLEM STATEMENT

Scalar ES contemplates the real-time control problem in which the aim is to find the maximum (or minimum) of the output $y \in \mathbb{R}$ of an unknown nonlinear static map $Q(\theta)$ by modifying the input $\theta \in \mathbb{R}$. In this section, we assume there is an arbitrarily large *time-varying and known* delay $D(t) \geq 0$ in the actuation path or measurement system such that the map’s output can be denoted by

$$y(t) = Q(\theta(t - D(t))). \quad (1)$$

For the sake of clarity, we assume that our system is output-delayed in the following presentation, according to the block diagram of Figure 1. However, the results in this paper can be directly generalized to the input-delay case once any input time-delay can be moved to the output of the static map. The general setup when input delays $D_{in}(t)$ and output delays $D_{out}(t)$ occur concomitantly can also be handled, by assuming that the total delay to be neutralized would be $D(t) = D_{in}(t) + D_{out}(t)$, with $D_{in}(t), D_{out}(t) \geq 0$.

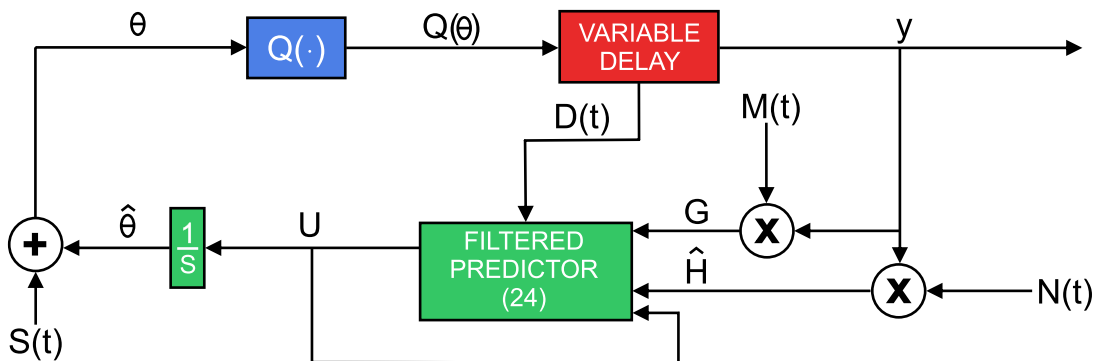


FIGURE 1. Basic block diagram of the extremum seeking control system with nonconstant delays.

Remark 1: The term “nonconstant delays” in this paper means variable delays which may depend exclusively on the time variable $D(t)$, exclusively on the state variable $D(\theta(t))$ or simultaneously on both $D(t, \theta(t))$. All of these three cases are handled in the manuscript. For the sake of clarity, we start the presentation with the simplest case of time-varying delays $D(t)$ and, then, we move forward to the more complicated cases involving state-dependent delays $D(\theta(t))$ in Section VII as well as simultaneous time- and state-dependent delays $D(t, \theta(t))$ in Section VIII.

A. LOCALLY QUADRATIC MAPS

Without loss of generality, let us consider the maximum seeking problem such that the maximizing value of θ is denoted by θ^* . For the sake of simplicity, we also assume that the nonlinear map is at least locally quadratic:

$$Q(\theta) = y^* + \frac{H}{2}(\theta - \theta^*)^2, \quad (2)$$

where besides the constants $\theta^* \in \mathbb{R}$ and $y^* \in \mathbb{R}$ being unknown, the scalar $H < 0$ is the unknown Hessian of the static map. By plugging (2) into (1), we obtain the *quadratic static map with delay* of interest:

$$y(t) = y^* + \frac{H}{2}(\theta(t - D(t)) - \theta^*)^2. \quad (3)$$

B. BASIC CONDITIONS ON TIME-VARYING DELAYS

The basic conditions on the arbitrarily large time-varying delay $D(t)$ are listed below [16, Chapter 6]:

- (a) The delay function $D(t)$ is strictly positive (technical condition ensuring that the state space of the input dynamics can be defined);
- (b) The delay function $D(t)$ is uniformly bounded from above, but there is no restriction in the amplitude of its constant upper bound;
- (c) The delay-rate function $\dot{D}(t)$ is strictly smaller than 1, i.e., the delay may increase at a rate smaller than 1;
- (d) The delay-rate function $\dot{D}(t)$ is uniformly bounded from below (by a possibly negative finite constant), i.e., the delay may decrease at a uniformly bounded rate;
- (e) The delay $D(t)$ is not periodic in $T = 2\pi/\omega$ such that

$$D_{av}(t) = D(t), \quad (4)$$

where $\omega > 0$ is the frequency of the dither signals in the ES loop. Any other frequency (different of ω) for a periodic time-varying delay is allowed.

Considering the continuously differentiable functions of the *delay time* $\phi(t)$ and *prediction time* $\phi^{-1}(t)$, defined as

$$\phi(t) := t - D(t), \quad (5)$$

$$\phi^{-1}(t) := t + D(t), \quad (6)$$

and satisfying the conditions

$$\phi(t) \leq t, \quad \frac{d}{dt}\phi(t) \geq 0, \quad \forall t \geq 0, \quad (7)$$

the following assumption can be assumed.

Assumption 1: The output time-varying delay $D(t)$ fulfills

$$-\infty < \underline{d} < \dot{D}(t) < \bar{d} < 1, \quad (8)$$

where $\underline{d}$ and $\bar{d}$ are constants. $\lrcorner$

This is a natural consequence of (5) and (6). In addition, from *Assumption 1* it follows that $\phi(t)$ is strictly increasing. Note that $\phi^{-1}(t)$ is the inverse function of $\phi(t)$.

C. SYSTEM AND SIGNALS

Let $\hat{\theta}$ be the estimate of θ^* and

$$\tilde{\theta}(t) = \hat{\theta}(t) - \theta^* \quad (9)$$

be the *estimation error*. From Figure 1 and noting that $\dot{\hat{\theta}}(t) = \dot{\tilde{\theta}}(t)$, the *error dynamics* can be written as

$$\dot{\tilde{\theta}}(t - D(t)) = U(t - D(t)), \quad (10)$$

where $U(t)$ is the control signal. Additionally, we have

$$G(t) = M(t)y(t), \quad \theta(t) = \hat{\theta}(t) + S(t), \quad (11)$$

with the dither signals being given by [18]:

$$S(t) = a \sin(\omega(t + D(t))), \quad M(t) = \frac{2}{a} \sin(\omega t), \quad (12)$$

with nonzero perturbation amplitude a and frequency ω . The signal

$$\hat{H}(t) = N(t)y(t) \quad (13)$$

is applied to obtain an estimate of the unknown Hessian H , where the demodulating signal $N(t)$ is given by

$$N(t) = -\frac{8}{a^2} \cos(2\omega t). \quad (14)$$

In [5], it was proved that

$$\frac{1}{T} \int_0^T N(\sigma)y d\sigma = H, \quad T = 2\pi/\omega, \quad (15)$$

if a quadratic map as in (2) is considered. In other words, its average version is $\hat{H}_{av} = (Ny)_{av} = H$.

IV. PREDICTOR FEEDBACK WITH HESSIAN'S ESTIMATE

By using the averaging analysis and from (4), we can verify that the average version of the signal $G(t)$ in (11) is given by

$$G_{av}(t) = H\tilde{\theta}_{av}(t - D(t)). \quad (16)$$

From (10), the following average models can be obtained

$$\dot{\tilde{\theta}}_{av}(t - D(t)) = U_{av}(t - D(t)), \quad (17)$$

$$\dot{G}_{av}(t) = H U_{av}(t - D(t)), \quad (18)$$

where $U_{av} \in \mathbb{R}$ is the resulting average control for $U \in \mathbb{R}$.

In order to motivate the predictor feedback design, the idea here is to compensate for the delay by feeding back the future state $G(t + D)$, or $G_{av}(t + D)$ in the equivalent average system. Given any stabilizing gain $k > 0$ for the undelayed system, our wish is to have a control that achieves

$$U_{av}(t) = k G_{av}(t + D(t)), \quad \forall t \geq 0, \quad (19)$$

and it appears to be non implementable since it requires future values of the state. However, by applying the variation of constants formula to (18) we can express the future state [16, Chapter 6] as

$$G_{av}(t + D(t)) = G_{av}(t) + H \int_{t-D(t)}^t \frac{U_{av}(\sigma)}{\phi'(\phi^{-1}(\sigma))} d\sigma, \quad (20)$$

which gives the future state $G_{av}(t+D)$ in terms of the average control signal $U_{av}(\sigma)$ from the past window $[t - D(t), t]$. To obtain (20), we have used the basic differentiation rule of the inverse of a function as follows

$$\frac{d}{d\sigma} \phi^{-1}(\sigma) = \frac{1}{\phi'(\phi^{-1}(\sigma))},$$

where ϕ' denotes the time derivative of $\phi(\cdot)$.

Plugging (20) into (19) yields the following feedback law

$$U_{av}(t) = k \left[G_{av}(t) + H \int_{t-D(t)}^t \frac{U_{av}(\sigma)}{\phi'(\phi^{-1}(\sigma))} d\sigma \right], \quad \forall t \geq 0. \quad (21)$$

Hence, from (20) and (21), the average feedback law (19) can be obtained indeed as desired. Consequently,

$$\dot{\tilde{\theta}}_{av}(t) = kG_{av}(t + D(t)), \quad \forall t \geq 0. \quad (22)$$

Therefore, from (16), one has

$$\frac{d\tilde{\theta}_{av}(t)}{dt} = kH\tilde{\theta}_{av}(t), \quad \forall t \geq D(t), \quad (23)$$

with an exponentially attractive equilibrium $\tilde{\theta}_{av}^e = 0$, since $k > 0$ in the control design and $H < 0$ by assumption.

Along of the remaining sections of the paper, we show that the control objectives can still be achieved if a simple modification of the above basic predictor-based controller, which employs a low-pass filter [29], is applied. In this case, we propose the next averaging-based predictor feedback in order to compensate the delay

$$U(t) = \frac{c}{s+c} \left\{ k \left[G(t) + \hat{H}(t) \int_{t-D(t)}^t \frac{U(\sigma)}{\phi'(\phi^{-1}(\sigma))} d\sigma \right] \right\}, \quad (24)$$

or, equivalently, in its state-space representation:

$$\dot{U}(t) = -cU(t) + ck \left[G(t) + \hat{H}(t) \int_{t-D(t)}^t \frac{U(\sigma)}{\phi'(\phi^{-1}(\sigma))} d\sigma \right], \quad (25)$$

$\forall t \geq 0$, where $c > 0$ is sufficiently large.

The predictor feedback is of the form of a low-pass filter of the non average version of (21). This low pass filtering is particularly required in the stability analysis when the averaging theorem in infinite dimensions [24] is invoked. Basically, this modification allows us to manipulate input-output delays as distributed state delays.

V. PREDICTOR NUMERICAL IMPLEMENTATION-APPROXIMATION FOR TIME-VARYING DELAYS

Apparently it may seem difficult to numerically implement the predictor (24). Thus, in the next sub-sections it will be shown two simplified methods of performing this implementation. Alternative implementations and approximations of predictor feedback for delay systems can also be find in [30].

A. FIRST METHOD

Considering equations (5) and (18), we can write

$$\dot{G}_{av}(t) = HU_{av}(\phi(t)). \quad (26)$$

The prediction of the state at the time when the current control will have an effect on the state is defined as

$$P_{av}(t) = G_{av}(t + D(\tau(t))), \quad (27)$$

where the prediction time is given by

$$\tau(t) = t + D(\tau(t)), \quad (28)$$

which is derived from the inversion of the time variable $t \rightarrow t - D(t)$ in $t \rightarrow t + D(\tau(t))$.

In order to utilize (28) instead of t , one rewrites (26) as

$$\dot{G}_{av}(\tau(t)) = HU_{av}(t), \quad (29)$$

and rewriting (29), we have:

$$\frac{dG_{av}(\tau(t))}{dt} = HU_{av}(t) \frac{d\tau(t)}{dt}. \quad (30)$$

Moreover, deriving equation (28), we get

$$\frac{d\tau(t)}{dt} = 1 + D'(\tau(t)) \frac{d\tau(t)}{dt}, \quad (31)$$

$$\frac{d\tau(t)}{dt} = \frac{1}{1 - D'(\tau(t))}. \quad (32)$$

Thus, (27) can be rewritten as

$$\frac{dP_{av}(t)}{dt} = HU_{av}(t) \frac{1}{1 - D'(\tau(t))}. \quad (33)$$

Integrating (31) and (33) for all $t - D(t) \leq \xi \leq t$, we obtain the following predictor [17]:

$$P_{av}(t) = G_{av}(t) + H \int_{t-D(t)}^t U_{av}(\xi) \frac{1}{1 - D'(\sigma(\xi))} d\xi, \quad (34)$$

$$\tau(t) = t + \int_{t-D(t)}^t \frac{1}{1 - D'(\tau(\xi))} d\xi, \quad (35)$$

Finally, the non average version for the predictor-based control law is given by

$$U(t) = \frac{c}{s+c} \left\{ k \left[G(t) + \hat{H}(t) \int_{t-D(t)}^t U(\xi) \frac{1}{1 - D'(\tau(\xi))} d\xi \right] \right\}. \quad (36)$$

Now, for example, using the point-to-point integration rule is easy to calculate (36) numerically, as described below.

Computation of the predictor signal: When simulating the predictor feedback controller (34)–(36), at each time step

the ODE for the system (26) must be solved (using, for example, a simple Euler scheme) and the length of the delay must be computed (for example, as the integer part of $N(i) = \frac{D(i)}{h}$, say $\bar{N}(i)$, where h is the discretization step). The predictor is then computed by integrating simultaneously the two integral relations (34) and (35) at each time step, using a numerical integration scheme. For instance, with the left endpoint rule of integration we get

$$P(i) = G(i) + h\hat{H}(i) \sum_{k=i-\bar{N}(i)}^{i-1} U(k) \frac{1}{1 - D'(\tau(k))}, \quad (37)$$

$$\tau(i) = i + h \sum_{k=i-\bar{N}(i)}^{i-1} \frac{1}{1 - D'(\tau(k))}. \quad (38)$$

B. SECOND METHOD

In [31], the authors consider the following predictor

$$U_{av}(t) = k \left[G_{av}(t) + H \int_{t-D(t)}^t U_{av}(s) ds \right], \quad (39)$$

for systems of the type $\dot{G}_{av} = H U_{av}(t - D(t))$, where H was originally assumed constant. Note that this controller does not exactly match the predicted system state on a time-horizon $D(t)$. Indeed, using the variation of constants formula, $\forall t \geq 0$:

$$G_{av}(t + D(t)) = G_{av}(t) + H \int_{t-D(t)}^t U_{av}(s + D(t) - D(s)) ds. \quad (40)$$

However, the integral in this prediction may not be implementable as it is not necessarily causal (in details, this is the case when there exists $s \in [t - D(t), t]$ such that $s - D(s) \geq t - D(t)$, i.e., when the delay $D(t)$ is suddenly high and the signal received at time t is older than those previously received) while the one employed in (39) always is [31].

Further, even if one can implement this prediction, the involved integral can be approximated by the one used in (39) if $D(t) - D(s) \approx 0$ for most instant t , i.e., under the assumption that the variations of the delay $\dot{D}(t)$ are sufficiently small in average. As this assumption is the one which is required in the following Theorem 1 to robustly compensate the delay, we rather use the prediction form (39) which is always causal and easier to implement.

The implementation of the proposed predictor starts by dividing the integral (39) into two parts, according to the following equation

$$\int_{t-D(t)}^t U_{av}(s) ds = \int_{t-D(t)}^0 U_{av}(s) ds + \int_0^t U_{av}(s) ds, \quad (41)$$

$$\int_{t-D(t)}^t U_{av}(s) ds = - \int_0^{t-D(t)} U_{av}(s) ds + \int_0^t U_{av}(s) ds. \quad (42)$$

Solving to integral below

$$\int_0^{t-D(t)} U_{av}(s) ds, \quad (43)$$

taking into account

$$s = t - D(t), \quad ds = dt - \dot{D}dt, \quad (44)$$

we get $t = D(t)$ for $s = 0$ and $t = t$ for $s = t - D(t)$. Substituting the above conditions at the limits of integration of the equation (43), we have

$$\int_{D(t)}^t U_{av}(t - D(t)) [dt - \dot{D}dt] \quad (45)$$

$$= \int_{D(t)}^t (1 - \dot{D}) U_{av}(t - D(t)) dt \quad (46)$$

$$= \int_0^t (1 - \dot{D}) U_{av}(t - D(t)) dt, \quad (47)$$

being valid if the conditions $\dot{D} \leq 1$ and $1 - \dot{D} \geq 0$ are satisfied and considering that

$$U_{av}(t - D(t)) = 0, \quad t \in [0, D(t)]. \quad (48)$$

Thus, the non-average version (39) could be rewritten as:

$$U(t) = k \left[G(t) + \hat{H}(t) \int_0^t (1 - \dot{D}(\sigma)) U(\sigma - D(\sigma)) d\sigma \right]. \quad (49)$$

Thus, instead of implementing (24), we could simply adopt the filtered version of (49).

VI. STABILITY ANALYSIS

Let us introduce the following non-obvious choice for the state of the transport equation [16, Chapter 6]:

$$u(x, t) = U(\phi(t + x(\phi^{-1}(t) - t))). \quad (50)$$

From (50), we get boundary values

$$u(0, t) = U(\phi(t)), \quad (51)$$

$$u(1, t) = U(t). \quad (52)$$

This yields to the following system

$$\dot{\theta}(t - D) = u(0, t), \quad (53)$$

$$u_t(x, t) = \pi(x, t) u_x(x, t), \quad x \in [0, 1], \quad (54)$$

$$u(1, t) = U(t), \quad (55)$$

where the speed of propagation of the transport equation is given by

$$\pi(x, t) = \frac{1 + x \left(\frac{d(\phi^{-1}(t))}{dt} - 1 \right)}{\phi^{-1}(t) - t}. \quad (56)$$

The equation (56) is obtained simply by calculating $\frac{u_t(x, t)}{u_x(x, t)}$ considering $u(x, t)$ in (50). Hence, the following condition can be assumed.

Assumption 2: The function (56) is strictly positive and uniformly bounded from below and from above by finite constants. Additionally, we assume $\pi(x, t)$ to be non periodic on $T = 2\pi/\omega$, so that $\pi_{av}(x, t) = \pi(x, t)$ is satisfied. $\square$

Moreover, we have

$$\pi(0, t) = \frac{1}{\phi^{-1}(t) - t} \geq \pi_0^* > 0, \quad (57)$$

where

$$\pi_0^* = \frac{1}{\sup_{\delta \geq \phi^{-1}(0)} (\delta - \phi(\delta))}. \quad (58)$$

Furthermore, we can write

$$\phi^{-1}(t) - t > 0, \quad (59)$$

which leads to the *delay-time* (5) and *prediction-time* (6) being both positive and uniformly bounded functions.

Now, rewriting the control law (24) in terms of $u(x, t)$, we get

$$u(1, t) = \frac{c}{s + c} \left\{ k \left[G(t) + \hat{H}(t) \int_0^1 u(\sigma, t) (\phi^{-1}(t) - t) d\sigma \right] \right\}. \quad (60)$$

The next initial condition is further assumed

$$u_0(x) = u(x, 0) = U(\phi(\phi^{-1}(0)x)), \quad x \in [0, 1], \quad (61)$$

with $\tilde{\theta}_0 = \tilde{\theta}(-D(0))$ and considering the following change of variables

$$\tilde{\vartheta}(t) = \tilde{\theta}(t - D(t)), \quad \tilde{\vartheta}_{av}(t) = \tilde{\theta}_{av}(t - D(t)), \quad (62)$$

the average version of system (53)–(55) can be directly written as

$$\dot{\tilde{\vartheta}}_{av}(t) = u_{av}(0, t), \quad (63)$$

$$\partial_t u_{av}(x, t) = \pi(x, t) \partial_x u_{av}(x, t), \quad (64)$$

$$u_{av}(1, t) = U_{av}(t), \quad (65)$$

or

$$\dot{\tilde{\vartheta}}_{av}(t) = u_{av}(0, t), \quad (66)$$

$$\partial_t u_{av}(x, t) = \pi(x, t) \partial_x u_{av}(x, t), \quad x \in [0, 1], \quad (67)$$

$$\begin{aligned} \frac{du_{av}(1, t)}{dt} &= -cu_{av}(1, t) \\ &+ ckH \left[\tilde{\vartheta}_{av}(t) + \int_0^1 u_{av}(\sigma, t) (\phi^{-1}(t) - t) d\sigma \right] \end{aligned} \quad (68)$$

Therefore, the next theorem for the average system can be guaranteed.

Theorem 1: Consider the closed-loop system consisting of the plant (63)–(65), the controller (60) and that Assumption 1 and Assumption 2 hold. There exist positive constants α_1, α_2, μ (independent of ϕ) such that the closed-loop system (63)–(65) is locally exponentially stable and the solution satisfies, $\forall t \geq 0$:

$$\begin{aligned} &\left[\left| \tilde{\theta}_{av}(\phi(t)) \right|^2 + [U_{av}(t)]^2 + \int_{\phi(t)}^t U_{av}^2(\xi) d\xi \right] \\ &\leq \frac{\alpha_1}{\alpha_2} e^{-\mu t} \left[\left| \tilde{\theta}_{av}(\phi(0)) \right|^2 + [U_{av}(0)]^2 + \int_{\phi(0)}^0 U_{av}^2(\xi) d\xi \right] \end{aligned} \quad (69)$$

Proof: Consider the infinite-dimensional backstepping transformation of transport PDE state [32]

$$\begin{aligned} w(x, t) &= u_{av}(x, t) \\ &- kH \left[\tilde{\vartheta}_{av}(t) + \int_0^x u_{av}(\sigma, t) (\phi^{-1}(t) - t) d\sigma \right] \end{aligned} \quad (70)$$

which maps the system (66)–(69) into the *target system* [16], [32]:

$$\dot{\tilde{\vartheta}}_{av}(t) = kH \tilde{\vartheta}_{av}(t) + w(0, t), \quad (71)$$

$$w_t(x, t) = \pi(x, t) w_x(x, t), \quad x \in [0, 1], \quad (72)$$

$$w(1, t) = -\frac{1}{c} \partial_t u_{av}(1, t). \quad (73)$$

In particular, the inclusion of the dynamic boundary condition [29] in (73) due to the inclusion of the low-pass filter in (24) and (25) was not reported in the references [16], [32] for the PDE representation of time-varying delayed systems.

The system (71)–(73) is a standard cascade configuration

$$w \rightarrow \tilde{\vartheta}_{av}. \quad (74)$$

Now, let us consider the term $w(1, t)$. Deriving (70) with respect to the spatial variable x and replacing $\partial_t u_{av}(x, t) = \pi(x, t) \partial_x u_{av}(x, t)$ and $w_t(x, t) = \pi(x, t) w_x(x, t)$ – see (67) and (72) – with $x = 1$, it is easily seen that

$$w_t(1, t) = \partial_t u_{av}(1, t) - kH \pi(1, t) u_{av}(1, t) [\phi^{-1}(t) - t], \quad (75)$$

where $\partial_t u_{av}(1, t) = \dot{U}_{av}(t)$. Equivalently, we can obtain

$$w_t(1, t) = \partial_t u_{av}(1, t) - kH \bar{\pi}^* u_{av}(1, t), \quad \bar{\pi}^* := \frac{d(\phi^{-1}(t))}{dt}, \quad (76)$$

by substituting the speed of propagation of the transport equation $\pi(x, t)$ in (56) with $x = 1$ into (75). Equation (76) is a more convenient expression, which will be used in the stability analysis since $\bar{\pi}^* = \frac{d(\phi^{-1}(t))}{dt} = 1 + \dot{D}(t)$ is upper bounded such that $|\bar{\pi}^*| \leq 2$, according to (6) and Assumption 2.

The inverse of (70) is given by

$$\begin{aligned} u_{av}(x, t) &= w(x, t) + kH e^{kHx(\phi^{-1}(t)-t)} \tilde{\vartheta}_{av}(t) \\ &+ kH \int_0^x e^{kH(x-\sigma)(\phi^{-1}(t)-t)} w(\sigma, t) (\phi^{-1}(t) - t) d\sigma. \end{aligned} \quad (77)$$

Plugging (73) and (77) into (76) with $x = 1$, we obtain

$$\begin{aligned} w_t(1, t) &= -cw(1, t) - kH \bar{\pi}^* w(1, t) \\ &- k^2 H^2 \bar{\pi}^* e^{kH(\phi^{-1}(t)-t)} \tilde{\vartheta}_{av}(t) \\ &- k^2 H^2 \bar{\pi}^* \int_0^1 e^{kH(1-\sigma)(\phi^{-1}(t)-t)} w(\sigma, t) (\phi^{-1}(t) - t) d\sigma. \end{aligned} \quad (78)$$

In [26], the term $\frac{a}{2} \int_0^1 (1+x) w^2(x, t) dx$ was assumed in the Lyapunov-Krasovskii functional. The time-derivative of

this term lead us to $\int_0^1 (1+x)\pi(x, t)w(x, t)w_x(x, t)dx$. In the proof of our previous conference version [26], we have used the following inequality

$$\int_0^1 (1+x)\pi(x, t)w(x, t)w_x(x, t)dx \leq \Pi \int_0^1 (1+x)w(x, t)w_x(x, t)dx, \quad (79)$$

with $\Pi \geq \pi(x, t) > 0$. However, the inequality (79) is not true since both w and w_x can be either positive or negative.

In this context and unlike [26], we now consider the following Lyapunov-Krasovskii functional [32]:

$$V(t) = \frac{\tilde{\vartheta}_{av}^2(t)}{2} + \frac{a}{2} \int_0^1 e^{bx} w^2(x, t) dx + \frac{1}{2} w^2(1, t), \quad (80)$$

where the parameters $a > 0$ and $b > 0$ are to be chosen later. With some abuse of notation, the constant a in (80) is not the amplitude of the perturbation signals given in (12).

From Assumption 2, we have that $\pi(0, t)$ and $\pi(1, t)$ are lower and upper bounded by $0 < \pi_0^* \leq \pi(0, t)$ and $0 < \pi(1, t) \leq \pi_1^*$, with positive constants π_0^* and π_1^* , the former defined in (58). Reminding that $w_t(x, t) = \pi(x, t)w_x(x, t)$ in (72) and applying the integration by parts to the integral term, the time derivative of (80) along of (71)–(73) can be written by

$$\begin{aligned} \dot{V}(t) &= kH \tilde{\vartheta}_{av}^2(t) + \tilde{\vartheta}_{av}(t)w(0, t) \\ &\quad + \frac{a}{2} \int_0^1 \overbrace{e^{bx} \pi(x, t)}^u \underbrace{2w(x, t)w_x(x, t)dx}_{dv} \\ &\quad + w(1, t)w_t(1, t) \\ &= kH \tilde{\vartheta}_{av}^2(t) + \tilde{\vartheta}_{av}(t)w(0, t) \\ &\quad + \frac{a}{2} e^b \pi(1, t)w^2(1, t) - \frac{a}{2} \pi(0, t)w^2(0, t) \\ &\quad - \frac{a}{2} \int_0^1 [b\pi(x, t) + \pi_x(x, t)]e^{bx} w^2(x, t) dx \\ &\quad + w(1, t)w_t(1, t), \end{aligned} \quad (81)$$

such that

$$\begin{aligned} \dot{V}(t) &\leq kH \tilde{\vartheta}_{av}^2(t) + \frac{\tilde{\vartheta}_{av}^2(t)}{2a\pi_0^*} \\ &\quad - \frac{a}{2} \int_0^1 [b\pi(x, t) + \pi_x(x, t)]e^{bx} w^2(x, t) dx \\ &\quad + w(1, t) \left[w_t(1, t) + \frac{a}{2} e^b \pi_1^* w(1, t) \right], \end{aligned} \quad (82)$$

since $-\frac{a}{2} \pi(0, t)w^2(0, t) \leq -\frac{a}{2} \pi_0^* w^2(0, t)$ and, from Young's inequality, we have

$$\begin{aligned} \tilde{\vartheta}_{av}(t)w(0, t) &\leq \left| \tilde{\vartheta}_{av}(t) \right| |w(0, t)| \\ &\leq \frac{\tilde{\vartheta}_{av}^2(t)}{2a\pi_0^*} + \frac{a\pi_0^* w^2(0, t)}{2}. \end{aligned} \quad (83)$$

Reminding that $k > 0$ and $H < 0$, let us choose $a = -1/(kH\pi_0^*)$. Then,

$$\begin{aligned} \dot{V}(t) &\leq \frac{kH}{2} \tilde{\vartheta}_{av}^2(t) - \frac{a}{2} \int_0^1 [b\pi(x, t) + \pi_x(x, t)]e^{bx} w^2(x, t) dx \\ &\quad + w(1, t) \left[w_t(1, t) + \frac{a}{2} e^b \pi_1^* w(1, t) \right] \\ &= -\frac{\tilde{\vartheta}_{av}^2(t)}{2a\pi_0^*} - \frac{a}{2} \int_0^1 [b\pi(x, t) + \pi_x(x, t)]e^{bx} w^2(x, t) dx \\ &\quad + w(1, t) \left[w_t(1, t) + \frac{a}{2} e^b \pi_1^* w(1, t) \right]. \end{aligned} \quad (84)$$

Next, we observe that

$$\pi_x(x, t) = \frac{\frac{d(\phi^{-1}(t))}{dt} - 1}{\phi^{-1}(t) - t} \quad (85)$$

is a function of t only. Hence,

$$\begin{aligned} b\pi(x, t) + \pi_x(x, t) &= \frac{b \left[1 + x \left(\frac{d(\phi^{-1}(t))}{dt} - 1 \right) \right] + \frac{d(\phi^{-1}(t))}{dt} - 1}{\phi^{-1}(t) - t} \\ &= \frac{b - 1 + \frac{d(\phi^{-1}(t))}{dt} + bx \left(\frac{d(\phi^{-1}(t))}{dt} - 1 \right)}{\phi^{-1}(t) - t}. \end{aligned} \quad (86)$$

Since this is a linear function on x , it follows that it has a minimum at either $x = 0$ or $x = 1$, so we get

$$\begin{aligned} b\pi(x, t) + \pi_x(x, t) &\geq \frac{\min \left\{ b - 1 + \frac{d(\phi^{-1}(t))}{dt}, (b+1) \frac{d(\phi^{-1}(t))}{dt} - 1 \right\}}{\phi^{-1}(t) - t}. \end{aligned} \quad (87)$$

Next, we note that

$$\bar{\pi}^* := \frac{d(\phi^{-1}(t))}{dt} = \frac{1}{\phi'(\phi^{-1}(t))} \geq \frac{1}{\sup_{\delta \geq \phi^{-1}(0)} \phi'(\delta)} = \pi^*, \quad (88)$$

which yields

$$b\pi(x, t) + \pi_x(x, t) \geq \frac{\min \{b - 1 + \pi^*, (b+1)\pi^* - 1\}}{\phi^{-1}(t) - t}. \quad (89)$$

Choosing

$$b \geq (1 - \pi^*) \max \left\{ 1, \frac{1}{\pi^*} \right\}, \quad (90)$$

and reminding that $\frac{1}{\phi^{-1}(t)-t} \geq \pi_0^*$ from (57), we get

$$b\pi(x, t) + \pi_x(x, t) \geq \pi_0^* \beta^*, \quad (91)$$

where

$$\beta^* = \min \{b - 1 + \pi^*, (b+1)\pi^* - 1\} > 0. \quad (92)$$

So, returning to $\dot{V}$, we have that

$$\begin{aligned} \dot{V}(t) &\leq -\frac{1}{2a\pi_0^*} \tilde{\vartheta}_{av}^2(t) - \frac{a}{2} \pi_0^* \beta^* \int_0^1 e^{bx} w^2(x, t) dx \\ &\quad + w(1, t) \left[w_t(1, t) + \frac{a}{2} e^b \pi_1^* w(1, t) \right]. \end{aligned} \quad (93)$$

Now, consider (93) along with (78). With a completion of squares, we obtain

$$\begin{aligned} \dot{V}(t) &\leq -\frac{1}{4a\pi_0^*} \tilde{\vartheta}_{av}^2(t) - Y \int_0^1 e^{bx} w^2(x, t) dx \\ &\quad + a\pi_0^* \left| k^2 H^2 \bar{\pi}^* e^{kH(\phi^{-1}(t)-t)} \right|^2 w^2(1, t) \\ &\quad + X \left\| k^2 H^2 \bar{\pi}^* \left[\phi^{-1}(t) - t \right] e^{kH(1-\sigma)(\phi^{-1}(t)-t)} \right\|^2 w^2(1, t) \\ &\quad + \left[\frac{a}{2} e^b \pi_1^* + |kH| \bar{\pi}^* \right] w^2(1, t) - c w^2(1, t). \end{aligned} \quad (94)$$

To obtain (94), we have used:

$$\begin{aligned} &-w(1, t) \left\langle k^2 H^2 \bar{\pi}^* \left[\phi^{-1}(t) - t \right] e^{kH(1-\sigma)(\phi^{-1}(t)-t)}, w(\sigma, t) \right\rangle \\ &\leq |w(1, t)| \left\| k^2 H^2 \bar{\pi}^* \left[\phi^{-1}(t) - t \right] e^{kH(1-\sigma)(\phi^{-1}(t)-t)} \right\| \|w(t)\| \\ &\leq Y \|w(t)\|^2 \\ &\quad + X \left\| k^2 H^2 \bar{\pi}^* \left[\phi^{-1}(t) - t \right] e^{kH(1-\sigma)(\phi^{-1}(t)-t)} \right\|^2 w^2(1, t), \end{aligned} \quad (95)$$

where the first inequality is the Cauchy-Schwartz and the second is Young's, the notation $\langle \cdot, \cdot \rangle$ denotes the inner product in the spatial variable $\sigma \in [0, 1]$, on which both $[\phi^{-1}(t) - t] e^{kH(1-\sigma)(\phi^{-1}(t)-t)}$ and $w(\sigma, t)$ depend, and $\| \cdot \|$ denotes the L_2 norm in σ . In addition, we have also applied Young's inequality to show that

$$\begin{aligned} &-k^2 H^2 \bar{\pi}^* e^{kH(\phi^{-1}(t)-t)} \tilde{\vartheta}_{av}(t) w(1, t) \\ &\leq a\pi_0^* \left| k^2 H^2 \bar{\pi}^* e^{kH(\phi^{-1}(t)-t)} \right|^2 w^2(1, t) + \frac{1}{4a\pi_0^*} \left| \tilde{\vartheta}_{av}(t) \right|^2. \end{aligned} \quad (96)$$

Then, from (94), we arrive at

$$\begin{aligned} \dot{V}(t) &\leq -\frac{1}{4a\pi_0^*} \tilde{\vartheta}_{av}^2(t) - Y \int_0^1 e^{bx} w^2(x, t) dx \\ &\quad - (c - c^*) w^2(1, t), \end{aligned} \quad (97)$$

where

$$\begin{aligned} c^* &= a\pi_0^* \left| k^2 H^2 \bar{\pi}^* e^{kH(\phi^{-1}(t)-t)} \right|^2 \\ &\quad + X \left\| k^2 H^2 \bar{\pi}^* \left[\phi^{-1}(t) - t \right] e^{kH(1-\sigma)(\phi^{-1}(t)-t)} \right\|^2 \\ &\quad + \left[\frac{a}{2} e^b \pi_1^* + |kH| \bar{\pi}^* \right], \end{aligned} \quad (98)$$

for appropriate constants $X > 0$ and $Y > 0$. Consequently, from (97), if c is chosen such that $c > c^*$, we obtain

$$\dot{V}(t) \leq -\mu V(t), \quad (99)$$

for some $\mu > 0$. Hence, the closed-loop system is exponentially stable in the sense of the full state norm

$$\left(\left| \tilde{\vartheta}_{av}^2(t) \right|^2 + \int_0^1 w^2(x, t) dx + w^2(1, t) \right)^{1/2}, \quad (100)$$

i.e., in the transformed variable $(\tilde{\vartheta}_{av}, w)$.

From $\left(\left| \tilde{\vartheta}_{av}^2(t) \right|^2 + \int_0^1 u_{av}^2(x, t) dx + u_{av}^2(1, t) \right)^{1/2}$, we achieve the exponential stability in the sense of the norm L_2 for the original variables $\tilde{\theta}_{av}$ and u_{av} . In this case, it is necessary to show that there exist positive scalars α_1 and α_2 such that

$$\alpha_1 \Psi(t) \leq V(t) \leq \alpha_2 \Psi(t), \quad (101)$$

with $\Psi(t) = \left| \tilde{\vartheta}_{av}^2(t) \right|^2 + \int_0^1 u_{av}^2(x, t) dx + u_{av}^2(1, t)$, or equivalently,

$$\Psi(t) = \left| \tilde{\theta}_{av}(t - D(t)) \right|^2 + \int_{t-D(t)}^t U_{av}^2(\tau) d\tau + U_{av}^2(t). \quad (102)$$

This can be directly established by using (70), (77), (80) and applying the Cauchy-Schwartz inequality and other mathematical computations, as in the proof of [16, Theorem 2.1]. Thus, from (99), we obtain

$$\Psi(t) \leq \frac{\alpha_2}{\alpha_1} e^{-\mu t} \Psi(0), \quad (103)$$

which is equivalent to inequality (69). $\square$

VII. STATE-DEPENDENT DELAYS

In this section, we consider the plant (1)–(3) subject to a delay function that depends explicitly on the system state [33]:

$$y(t) = Q(\theta(t - D(\theta(t)))). \quad (104)$$

This problem is particularly important in distributed extremum seeking control of mobile robots or formation control with multi-agents, where the magnitude of the delay may depend on the distance of the robots/agents from the operator interface, or from the virtual leaders [34], [35].

One more interesting feature is that the delay function can even depend on unmeasured signals, but respecting the condition that at least the analytical expression (parametrization) for the delays is still known and the unmeasured signals can be estimated in the average sense. Without loss of generality, let us assume here that the delay function in (104) is:

$$D(\theta(t)) = Q'(\theta(t)) = H(\theta(t) - \theta^*), \quad (105)$$

with $Q(\theta)$ defined in (2). In this case, the delay function is exactly the expression of the gradient of the map we want to maximize.

Now, we conveniently redefine the estimation error introduced in (9) as

$$\tilde{\theta}(t) = \hat{\theta}(t - D(\theta(t))) - \theta^*, \quad (106)$$

and recalling that $\dot{\hat{\theta}}(t) = \dot{\hat{\theta}}(t - D(\theta(t))) = U(t - D(\theta(t)))$, we can write the associated error dynamics as

$$\dot{\tilde{\theta}}(t) = U(t - D(\theta(t))), \quad (107)$$

where $U(t)$ is the control signal.

Similarly to (1), but now assuming (104) and (106), the following average version of the signal $G(t)$ in (11) can be derived straightforwardly by

$$G_{av}(t) = H\tilde{\theta}_{av}(t), \quad (108)$$

From (107) and (108), the following average models can also be obtained

$$\dot{\tilde{\theta}}_{av}(t) = U_{av}(\phi(t)), \quad (109)$$

$$\dot{G}_{av}(t) = HU_{av}(\phi(t)), \quad (110)$$

$$\phi(t) = t - D(G_{av}(t)). \quad (111)$$

since $\theta(t - D(\theta(t))) = \hat{\theta}(t - D(\theta(t))) + S(t - D(\theta(t))) = \tilde{\theta}(t) + \theta^* + a \sin(\omega t)$ from the definitions in (11) and (106). Hence, $[\theta(t - D(\theta(t))) - \theta^*]_{av} = [\tilde{\theta}(t)]_{av}$ since $[a \sin(\omega t)]_{av} = 0$. By multiplying both sides of the former equation by H , from (105) and (108), we can write, with some abuse of notation, $[D(\theta(t))]_{av} = G_{av}(t) := D(G_{av}(t))$. The term $U_{av} : [\phi(t_0), \infty) \rightarrow \mathbb{R}, t \geq t_0 \geq 0, D \in C^1(\mathbb{R}; \mathbb{R}_+)$ is the average control of $U \in \mathbb{R}$.

Defining the prediction time as $\sigma(t) = \phi^{-1}(t)$ the inversion of the time variable $t \rightarrow t - D(G_{av}(t))$ in $t \rightarrow t + D(P_{av}(t))$, the following hold:

$$P_{av}(t) = G_{av}(P_{av}(t)), \quad (112)$$

$$\sigma(t) = t + D(P_{av}(t)), \quad (113)$$

where $P_{av}(t)$ is the associated predictor state.

Differentiating (112) and (113) and using (110), we arrive at [33]:

$$\frac{dG_{av}(\sigma(t))}{dt} = HU_{av}(t) \frac{d\sigma(t)}{dt}, \quad (114)$$

and

$$\dot{\sigma}(t) = \frac{1}{1 - H\nabla D(P_{av}(t))U_{av}(t)}, \quad (115)$$

with $\dot{D} = \nabla D \dot{G}_{av} = \nabla D H U_{av}$, ∇ being the gradient operator.

The following implicit integral relations [33] are derived for the predictor state and the prediction time by integrating (114) and (115) for all delay interval $\phi(t) \leq \Theta \leq t$:

$$P_{av}(\Theta) = G_{av}(t) + \int_{\phi(t)}^{\Theta} \frac{HU_{av}(s)}{1 - H\nabla D(P_{av}(s))U_{av}(s)} ds, \quad (116)$$

$$\sigma(\Theta) = t + \int_{\phi(t)}^{\Theta} \frac{1}{1 - H\nabla D(P_{av}(s))U_{av}(s)} ds. \quad (117)$$

Regarding the nominal feedback control gain $k > 0$ for the plant free of delays, the predictor feedback control law can be written as

$$U_{av}(t) = k \left[G_{av}(t) + \int_{\phi(t)}^t \frac{HU_{av}(s)}{1 - H\nabla D(P_{av}(s))U_{av}(s)} ds \right]. \quad (118)$$

The predictor feedback control law (118) is subject to the following feasibility condition [33]:

$$H\nabla D(P_{av}(t))U_{av}(t) < 1. \quad (119)$$

This is due to the fact that the prediction signal can be only generated if (116) and (117) are well-posed. If the condition (119) is violated, the control signal is directed toward the opposite direction to the plant.

Thus, the non-average version for the filtered predictor-based control law is given by

$$U(t) = \frac{c}{s + c} \{k [G(t) + \Gamma(t)]\}, \quad k > 0, \quad c > 0, \quad (120)$$

$$\Gamma(t) = \hat{H}(t) \int_{\phi(t)}^t \frac{U(\tau)}{1 - \hat{H}(\tau) \nabla D(G(\tau))U(\tau)} d\tau. \quad (121)$$

The stability proof can be established using an analogous average infinite-dimensional representation of the actuator state (63)–(65) with the propagation speed defined as

$$\pi(x, t) = \frac{1 + x(\dot{\sigma}(t) - 1)}{\sigma(t) - 1}. \quad (122)$$

The infinite-dimensional representation for the system (109)–(111) is

$$\dot{\tilde{\theta}}_{av}(t) = u_{av}(0, t), \quad (123)$$

$$\partial_t u_{av}(x, t) = \pi(x, t) \partial_x u_{av}(x, t), \quad x \in [0, 1], \quad (124)$$

$$\frac{du_{av}(1, t)}{dt} = -cu_{av}(1, t) + ckH \left[\tilde{\theta}_{av} + (\sigma(t) - t) \int_0^1 u_{av}(x, t) dx \right], \quad (125)$$

which can be mapped into the *target system* [33]:

$$\dot{\tilde{\theta}}_{av}(t) = kH\tilde{\theta}_{av}(t) + w(0, t), \quad (126)$$

$$w_t(x, t) = \pi(x, t)w_x(x, t), \quad x \in [0, 1], \quad (127)$$

$$w(1, t) = -\frac{1}{c} \partial_t u_{av}(1, t), \quad (128)$$

with the help of the following invertible backstepping transformation¹:

$$w(x, t) = u_{av}(x, t) - kH \left[\tilde{\theta}_{av}(t) + (\sigma(t) - t) \int_0^x u_{av}(\sigma, t) d\sigma \right]. \quad (129)$$

Considering the next Lyapunov-Krasovskii functional

$$V(t) = \frac{\tilde{\theta}_{av}^2(t)}{2} + \frac{a}{2} \int_0^1 e^{bx} w^2(x, t) dx + \frac{1}{2} w^2(1, t), \quad (130)$$

with appropriate constants $a > 0$ and $b > 0$, the proof of stability can be provided following the same procedure for time-varying delays. The main difference is that the delay time $\phi(t)$ depends on the state as well as the prediction time $\phi^{-1}(t)$ —here denoted by $\sigma(t)$. [See Appendix]

¹The infinite-dimensional representation of the predictor state is defined following the transformation of the actuator state (50), where $\phi(t)$ is now the state-dependent delay given by (111).

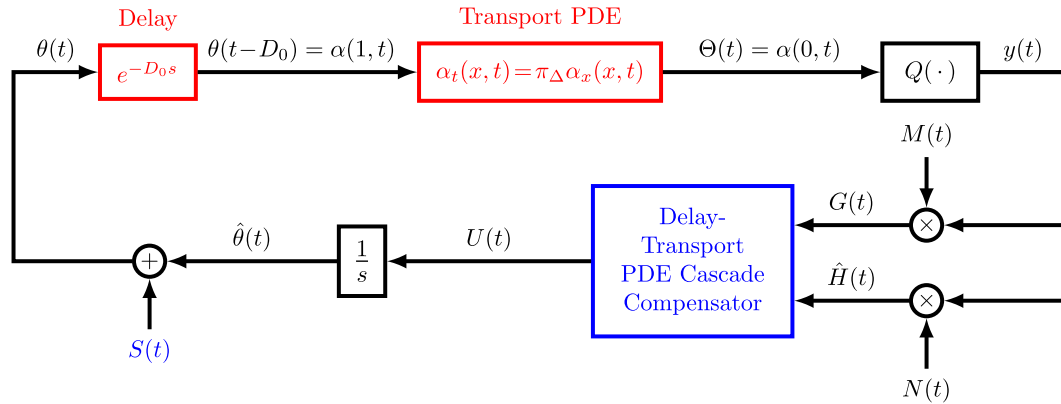


FIGURE 2. Block diagram for implementation of ES control design for the delay-PDE cascade (132)–(134) at the input of nonlinear convex maps $y = Q(\Theta)$. Such a representation encompasses simultaneous time- and state-dependent delays $D(t, \theta)$ in (131). Although the multiplicative perturbation signals $M(t)$ and $N(t)$ are the same of the classical ES designs [4], the additive dither $S(t)$ must be redesigned according to (150) and the application of an adequate boundary control law (149) for delay-PDE compensation is mandatory.

VIII. TIME- AND STATE-DEPENDENT DELAYS AS PDE-PDE CASCADES

In this section, we deal with cascades [16, Part V] of hyperbolic transport equations. We cope with an hyperbolic transport PDE (of variable convection speed) [36] with constant input delay. This is particularly interesting to represent time-varying delays $D(t)$ of Section III, state-dependent delays $D(\theta)$ in Section VII, or even simultaneous time- and state-dependent delays $D(t, \theta)$, as discussed in the following.

In this sense, we can represent a time- and state-dependent delay $D(t, \theta) \geq 0$ as [37], [38]:

$$D(t, \theta) = D_0 + \Delta(t, \theta), \quad (131)$$

where the scalar $D_0 > 0$ denotes the constant parcel of the delay and the term $\Delta(t, \theta) \in \mathbb{R}$ its variable portion.

Under this sort of delays (131) and from Figure 2, we can write the equations for the following delay-hyperbolic PDE cascade system [16, Part V]:

$$\alpha(0, t) = \Theta(t) \quad (132)$$

$$\partial_t \alpha(x, t) = \pi_\Delta(x, t) \partial_x \alpha(x, t), \quad x \in [0, 1] \quad (133)$$

$$\alpha(1, t) = \theta(t - D_0), \quad (134)$$

where $\alpha(x, t)$ is the infinite-dimensional state of the hyperbolic transport PDE with variable convection speed $\pi_\Delta(x, t) > 0$ used to represent $\Delta(t, \theta)$ in (131) and spatial domain defined without loss of generality by $x \in [0, 1]$. The boundary delay is denoted by D_0 being any arbitrary known constant given in (131). In (132), we conveniently denote the input of the map $y = Q(\Theta)$ by its inverse function $\Theta = Q^{-1}(y)$.

Consequently, we can rewrite (53)–(55) into the following delay-hyperbolic PDE dynamic cascade system:

$$\dot{\vartheta}(t) = u(0, t), \quad (135)$$

$$\partial_t u(x, t) = \pi_\Delta(x, t) \partial_x u(x, t), \quad x \in [0, 1] \quad (136)$$

$$u(1, t) = v(1, t), \quad (137)$$

$$\partial_t v(x, t) = \partial_x v(x, t), \quad x \in [1, 1 + D_0] \quad (138)$$

$$v(1 + D_0, t) = U(t), \quad (139)$$

where $U(t)$ is the overall system (control) input and (ϑ, u, v) is the state of the ODE-PDE-PDE cascade. From the transport PDE representation form for the input delay, we know that [16, Chapter 2]

$$v(x, t) = U(t + x - 1 - D_0), \quad x \in [1, 1 + D_0]. \quad (140)$$

Applying the backstepping transformations (with an appropriate prediction time $\sigma(t)$):

$$\begin{aligned} w_{av}(x, t) &= u_{av}(x, t) \\ &- k \left[H \tilde{\vartheta}_{av}(t) + (\sigma(t) - t) H \int_0^x u_{av}(\sigma, t) d\sigma \right], \\ x &\in [0, 1], \end{aligned} \quad (141)$$

$$\begin{aligned} \zeta_{av}(x, t) &= v_{av}(x, t) - kH \int_1^x v_{av}(\sigma, t) d\sigma \\ &- kH \int_0^1 u_{av}(\sigma, t) d\sigma - kH \vartheta_{av}(t), \\ x &\in [1, 1 + D_0], \end{aligned} \quad (142)$$

to the average version of (135)–(139), the resulting exponentially stable [16, Part V] average target system is

$$\dot{\vartheta}_{av}(t) = kH \vartheta_{av}(t) + w_{av}(0, t), \quad (143)$$

$$\partial_t w_{av}(x, t) = \pi_\Delta(x, t) \partial_x w_{av}(x, t), \quad (144)$$

$$w_{av}(1, t) = \zeta_{av}(1, t), \quad (145)$$

$$\partial_t \zeta_{av}(x, t) = \partial_x \zeta_{av}(x, t), \quad x \in [1, 1 + D_0] \quad (146)$$

$$\zeta_{av}(1 + D_0, t) = -\frac{1}{c} \partial_t v_{av}(1 + D_0, t). \quad (147)$$

From (142) and (147), one can obtain the expression for $\dot{U}_{av}(t) = -c\zeta_{av}(1 + D_0, t)$ such that

$$\begin{aligned} \dot{U}_{av}(t) = & -cv_{av}(1 + D_0, t) + ckH \int_1^{1+D_0} v_{av}(\sigma, t) d\sigma \\ & + ckH \int_0^1 u_{av}(\sigma, t) d\sigma + ckH \vartheta_{av}(t). \end{aligned} \quad (148)$$

Plugging (140) into (148), we can obtain the control law:

$$U(t) = \frac{c}{s+c} \left\{ k \left[G(t) + \hat{H}(t) \int_{t-D_0}^t U(\tau) d\tau + \hat{H}(t) \int_0^1 u(x, t) dx \right] \right\}, \quad (149)$$

for $c > 0$ sufficiently large.

As discussed in [23], the explicit solution for the trajectory generation problem [39, Chapter 12] under time- and state-dependent delays would be solved analogously to (12) as $S(t - D(t, \theta)) = a \sin(\omega t)$, leading us to

$$S(t) = a \sin(\omega(t + D(t, \theta))). \quad (150)$$

IX. AVERAGING FOR NONCONSTANT DELAYS

In Theorem 1, we proved that the average closed-loop system (53)–(55) with transport PDE for delay representation is exponentially stable. In [18]–[23] we have employed the averaging results for fixed delays. Nevertheless, for time-varying delays (as well as state-dependent delays $D(\theta)$ or simultaneous time- and state-dependent delays $D(t, \theta)$) there is no suitable averaging theorem such as in [24] and [25].

From [24], [25] and under the assumption of constant delays ($\dot{D}(t) \equiv 0$), we would be able to write the closed-loop system (10) and (25) into the form

$$\dot{\eta}(t) = f(t/\epsilon, \eta_t), \quad (151)$$

where $\eta(t) = [\tilde{\theta}(t - D), U(t)]^T$, $\eta_t(r) = \eta(t + r)$ for $-D \leq r \leq 0$, and f is an adequate continuous functional, such that the averaging theorem by [24], [25] (see Appendix of [23] for a compact version) can be straightforwardly applied, assuming $\omega = 1/\epsilon$. The gap here is very subtle: for instance, if the delay is time-varying ($\dot{D}(t) \neq 0$), we need to prove if the function $r(t)$, which is also time varying, would result in $-D_{max} \leq r(t) \leq 0$, with $D_{max} > 0$ being a constant upper bound for $D(t)$. This point may not seem of simple proof, and a formal demonstration was not given in the literature, which would involve the theory of semigroups for linear operators applied to PDEs [40].

In some particular cases, where the average of the variable term of the delay such as $\Delta(t, \theta)$ in (131) has average zero or even constant average ($\Delta_{av}(t, \theta_{av}) = 0$, $\Delta_{av}(t, \theta_{av}) = \text{constant}$), thus the cascade average system (144)–(145) becomes simply a cascade of two transport PDEs with fixed convection speeds, or equivalently, a string of constant delays for those the representation (151) would be valid and the infinite-dimensional averaging theory in [24] and [25] could still be employed.

Since the derivations of a completely novel averaging theorem for general variable delays is not the focus of the current paper, we leave this challenging and interesting generalization for a future contribution to be pursued.

If an appropriate averaging theorem existed, then one would be able to obtain the next theorem for time-varying delays (as well as an analogous version for the case of state-dependent delays).

Theorem 2: Consider the control system in Figure 1 with locally quadratic nonlinear map (1)–(3) under time-varying output delays satisfying Assumptions 1 and 2. There exists $c^* > 0$ such that, $\forall c \geq c^*$, $\exists w^*(c) > 0$ such that, $\forall \omega > \omega^*$, the delayed closed-loop system (10) and (25), with $G(t)$ in (11), $\hat{H}(t)$ calculated as in (13), and state $\tilde{\theta}(t - D(t))$, $U(\sigma)$, $\forall \sigma \in [t - D(t), t]$, has a unique locally exponentially stable periodic solution in t in the period $T = 2\pi/\omega$, denoted by $\tilde{\theta}^T(t - D(t))$, $U^T(\sigma)$, $\forall \sigma \in [t - D(t), t]$, satisfying, $\forall t \geq 0$:

$$\begin{aligned} & \left(\left| \tilde{\theta}^T(t - D(t)) \right|^2 + \left[U^T(t) \right]^2 + \int_{t-D(t)}^t \left[U^T(\tau) \right]^2 d\tau \right)^{1/2} \\ & \leq \mathcal{O}\left(\frac{1}{\omega}\right). \end{aligned} \quad (152)$$

Furthermore,

$$\limsup_{t \rightarrow +\infty} |\theta(t) - \theta^*| = \mathcal{O}(a + 1/\omega), \quad (153)$$

$$\limsup_{t \rightarrow +\infty} |y(t) - y^*| = \mathcal{O}(a^2 + 1/\omega^2). \quad (154)$$

For the time-varying delays case, the proof of asymptotic convergence to the extremum point (θ^*, y^*) , delineated by equations (153) and (154) is very close to that constructed in [23] for the case of constant delays, except for some intuitive changes. Please refer to Step 7 of the proof of Theorem 1 in [23] for more details.

X. SIMULATION RESULTS

Unlike most of the results in the literature which consider stabilization, tracking or disturbance rejection problems [41], [42], as discussed before, we are handling nonconstant time-delays in real-time optimization via ES.

Efforts in delay-compensation for ES feedback are motivated by the interest to address the lack of delay robustness identified by [18]–[23], showing that standard feedback laws for ES have a zero robustness margin to the introduction of a delay in the feedback loop—an arbitrarily small measurement delay or input delay results in closed-loop instability. In [18]–[23], only constant delays were considered.

In order to evaluate the proposed predictor-based ES approach for nonconstant delays compensation, let us assume the following quadratic static map

$$Q(\theta) = 5 - 0.1(\theta - 2)^2 \quad (155)$$

with an output delay $D(t) = 2 \sin(\omega_0 t) + D_0$ seconds, where $\omega_0 = 0.1$ rad/s and $D_0 = 5$. From (155), the extremum coordinates are given by $(\theta^*, y^*) = (2, 5)$ and the map's

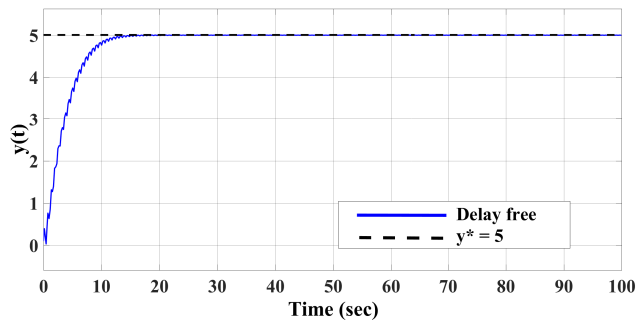


FIGURE 3. Output $y(t)$ without delays.

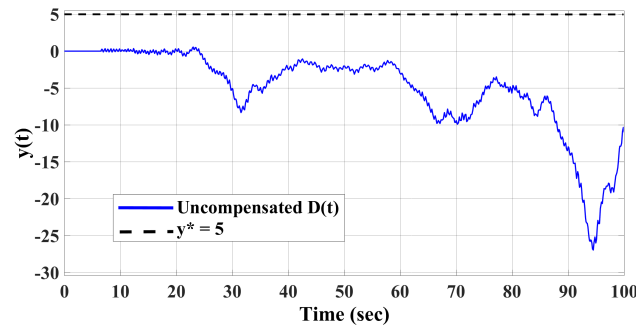


FIGURE 4. Time evolution of the output $y(t)$ under $D(t) = 2 \cos(0.1 t) + 5$ sec and employing the predictor for constant delays proposed in [18].

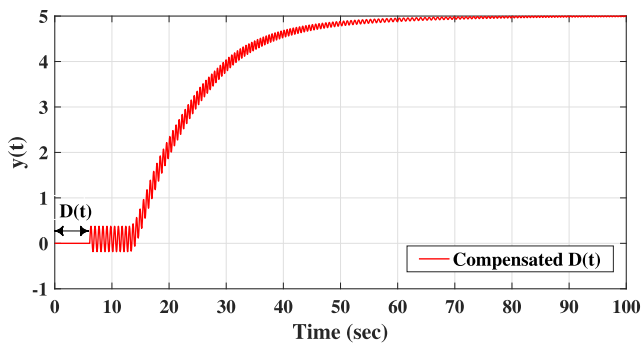


FIGURE 5. Time evolution of the output $y(t)$ under $D(t) = 2 \cos(0.1 t) + 5$ sec and employing the predictor for time-varying delays in (24).

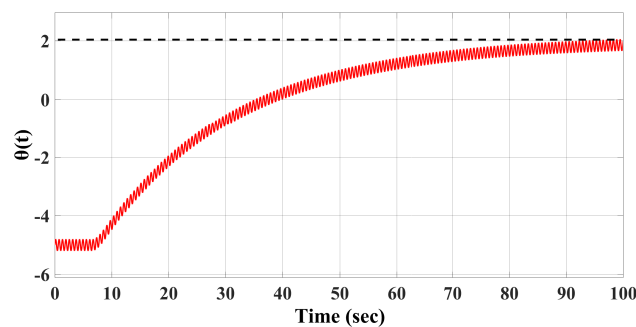


FIGURE 6. Input parameter $\theta(t)$ converging to the maximizer $\theta^* = 2$.

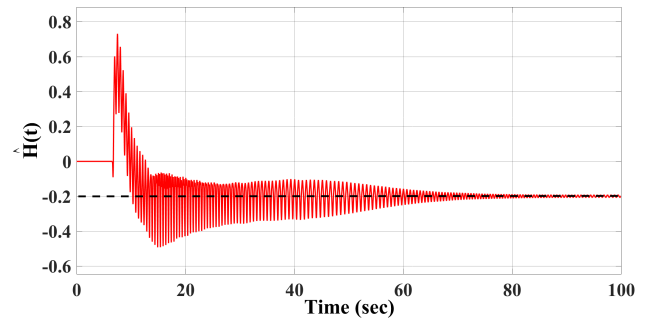


FIGURE 7. Time evolution of the Hessian estimate $\hat{H}(t)$ converging to $H = -0.2$.

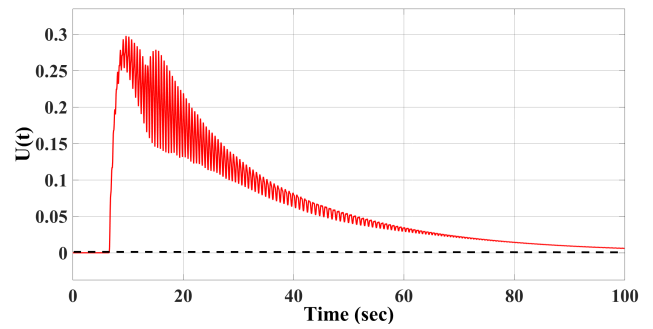


FIGURE 8. Control signal $U(t)$.

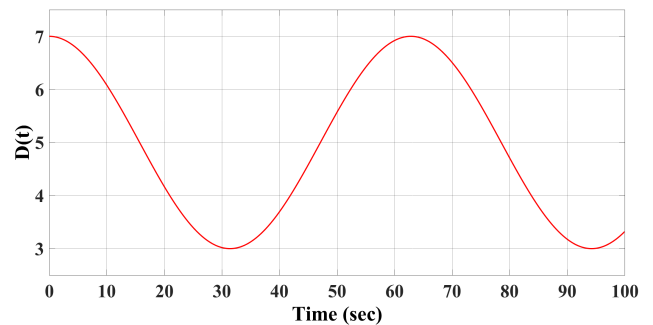


FIGURE 9. Time-varying nature of the delay $D(t) = 2 \cos(0.1 t) + 5$ in secs.

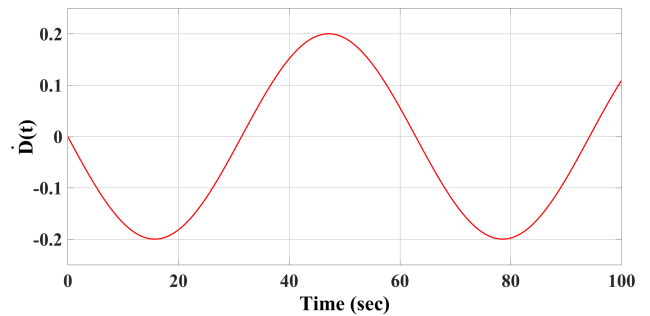
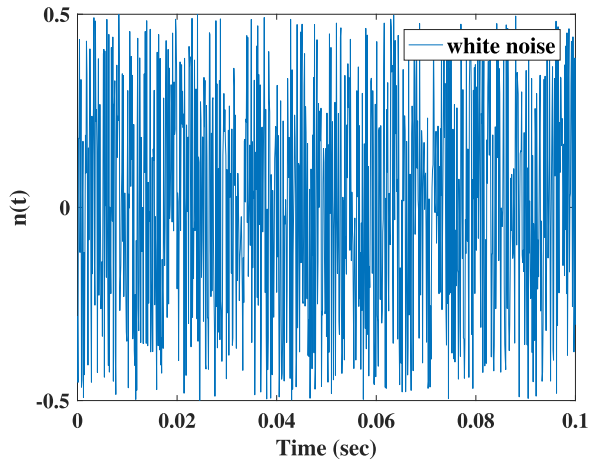
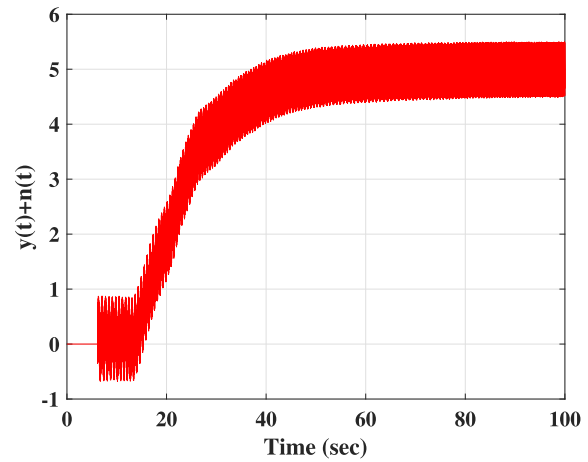
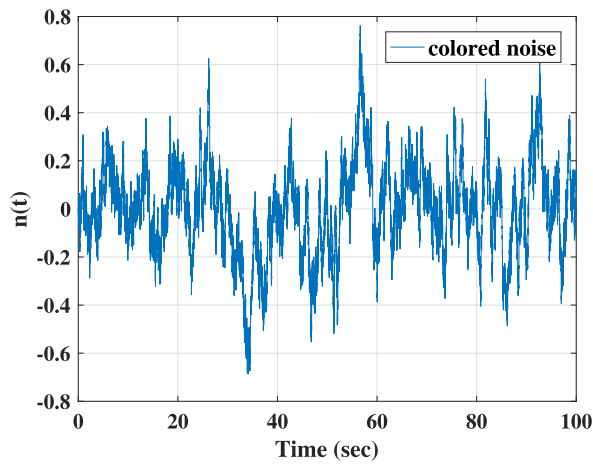
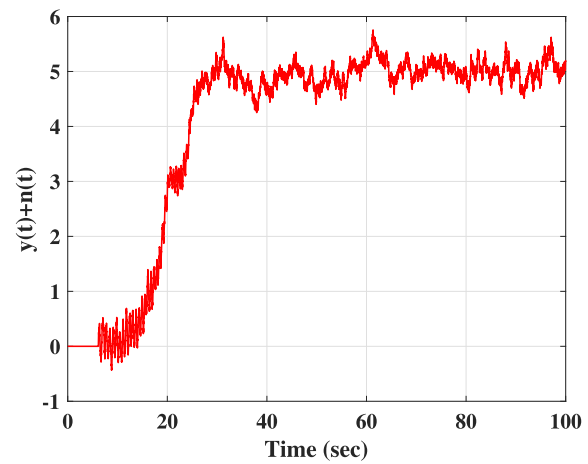


FIGURE 10. Time-derivative of the delay $\dot{D}(t) = -0.2 \sin(0.1 t)$.

(a) Zoom of the white noise signal $n(t)$ used in the numerical tests.(b) Output of the static map $y(t)$ corrupted by a white noise $n(t)$ converging to a neighborhood of $y^* = 5$.**FIGURE 11. Measurement Noise Effects: output signal corrupted by a white noise $y(t) + n(t)$.**(a) The colored noise signal $n(t)$ used in the numerical tests.(b) Output of the static map $y(t)$ corrupted by a colored noise $n(t)$ converging to a neighborhood of $y^* = 5$.**FIGURE 12. Measurement Noise Effects: output signal corrupted by a colored noise $y(t) + n(t)$. The colored noise is generated by filtering out the white noise employed in the plot of Figure 11(a) through the low-pass filter $H_0(s) = \frac{k}{s+c}$, with $k = 100$ and $c = 1$ rad/s.**

Hessian is chosen as $H = -0.2$. In the following, the numerical tests of the predictor (24) are carried out with $\hat{H}$ defined by (13) and considering $c = 20$. The other parameters for simulation were chosen as: $a = 0.2$, $\omega = 10$, $k = 0.2$ and initial condition $\theta(0) = -5$.

The Euler Method with step-size $h = 10^{-4}$ sec is used for numerical integration. Hence, all signals in the numerical tests are considered continuous since h is chosen sufficiently small. However, it is worth mention that any fast change (of a couple of seconds) within a time-range of hundred of seconds may give a false impression of a discontinuous behavior in the simulation plots.

The output $y(t)$ is shown in three different situations: (a) without delay (Figure 3); (b) in the presence of a time-varying output delay, but with predictor-feedback compensation proposed in [18] considering a nominal constant delay of $D_0 = 5$ seconds (Figure 4); and (c) with output delay and predictor-based compensator proposed in

equation (24) – see Figure 5. Although basic ES works well without delays, it goes unstable in the presence of delays if no compensation is applied or if an incorrect prediction scheme for constant delays [18] is implemented instead. The proposed predictor for time-varying delays fixes this.

In Figure 6, the input parameter θ is displayed, converging to θ^* . Figure 7 shows the estimate of unknown Hessian H , whereas Figure 8 presents the plot for the control signal $U(t)$. At last, the variable profile of the considered time-varying delay is displayed in Figure 9, whereas its derivative $\dot{D}(t) = -0.2 \sin(0.1 t)$ is illustrated in Figure 10.

Figures 11(a), 11(b) and Figures 12(a), 12(b) show the robustness and the control performance of the proposed extremum seeking approach in the presence of different types of measurement noise $n(t)$. In Figures 11(a) and 11(b) a white noise is considered, whereas a colored noise is employed in Figures 12(a) and 12(b). Under both adverse scenarios, the extremum search is preserved and the ultimate error

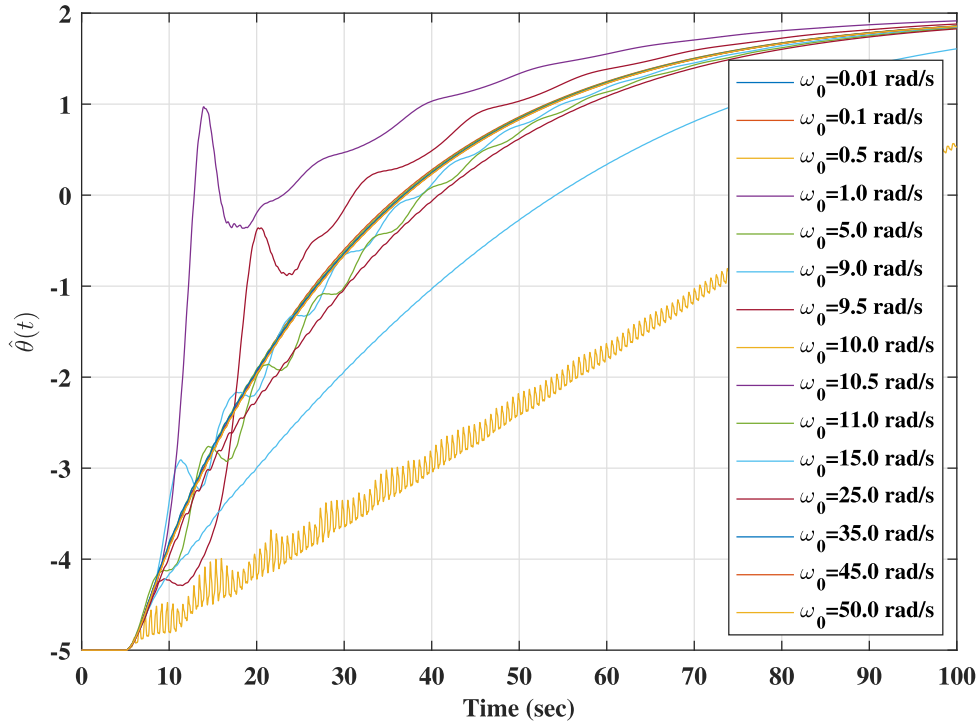


FIGURE 13. Time evolution of the estimate $\hat{\theta}(t)$ converging to $\theta^* = 2$ under different time-varying delay functions. Effects of the time-varying delay function $D(t) = 5 + 2\epsilon \sin(\omega_0 t)$ in the convergence of the algorithm for different frequencies $\omega_0 \in (0, 50]$ satisfying Assumption 1 with $\omega_0 < \frac{1}{2\epsilon}$ and $\epsilon = 0.01$. At the critical ratio of the frequencies $\omega/\omega_0 = 1$, a very slow undesired performance is generated since the delay function $D(t)$ becomes periodic in $T = 2\pi/\omega$ such that the average condition $D_{av}(t) = D(t)$ is violated.

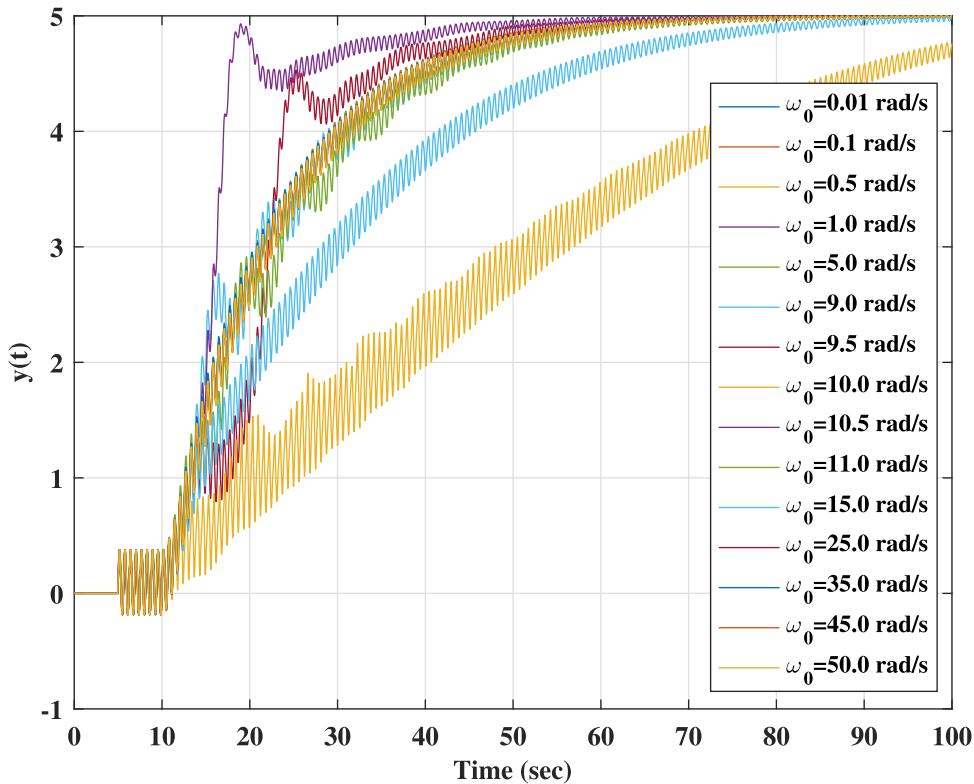


FIGURE 14. Time evolution of the output signal $y(t)$ converging to a neighborhood of the extremum point $y^* = 5$ under different time-varying delay functions. Effects of the time-varying delay function $D(t) = 5 + 2\epsilon \sin(\omega_0 t)$ in the convergence of the algorithm for different frequencies $\omega_0 \in (0, 50]$ satisfying Assumption 1 with $\omega_0 < \frac{1}{2\epsilon}$ and $\epsilon = 0.01$. At the critical ratio of the frequencies $\omega/\omega_0 = 1$, a very slow undesired performance is generated since the delay function $D(t)$ becomes periodic in $T = 2\pi/\omega$ such that the average condition $D_{av}(t) = D(t)$ is violated.

bound is kept within reasonable limits (convergence to a small residual set around the extremum $y^* = 5$).

Finally, Figures 13 and 14 present the effect of the time-varying delay function in the convergence of the algorithm for different time-varying delays $D(t) = 5 + 2\epsilon \sin(\omega_0 t)$ with $\omega_0 \in (0, 50]$ and $\epsilon = 0.01$. As $\omega_0 \rightarrow \omega$, the control system performance is severely affected, as expected. When $\omega = \omega_0$ the delay function $D(t)$ is periodic in $T = 2\pi/\omega$ such that the condition $D_{av}(t) = D(t)$ in (4) is violated, resulting in a very slow convergence of the ES algorithm to the desired value. Despite of that, the algorithm demonstrates some degree of robustness and continues the search for the extremum point.

XI. CONCLUSION

In this paper, we provided a new predictor feedback design for gradient-based extremum seeking in order to address arbitrarily large variable delays in the output (input) signals of an unknown locally quadratic static map. The predictor control law with distributed terms of the delay is developed with perturbation-based estimates for the Gradient and Hessian of the nonlinear map to be optimized. Numerical implementation aspects of the proposed predictor were also addressed. We present all the steps of the proof for local exponential convergence to a small neighborhood of the extremum point using analysis tools of backstepping transformation and averaging in infinite dimensions. The treatment of time- and state-dependent delays was also discussed as well as the possibility of designing extremum seeking boundary control laws by means of representative PDE-PDE cascades of hyperbolic types. Numerical results are performed to support the theoretical advances.

Along the manuscript, we have show how to properly sequence the steps of backstepping, Lyapunov functional analysis, and averaging to prove stability. We believe this theoretical pathway will guide future researchers who deal with extremum seeking under nonconstant delays.

The Newton version of the proposed algorithm is developed in [43]. Our preliminary results about extremum seeking for PDE-PDE cascades could be expanded to encompass more general cascades of PDEs from different families rather than first-order hyperbolic (transport) PDEs. Future research direction could also generalize the current results to unknown delays [44], [45] and investigate their application to experimental scenarios for mobile robotics or control over networks [33]–[35], transportation of materials [36], traffic control [46] and additive manufacturing and 3D printing [38].

APPENDIX ALTERNATIVE STABILITY PROOF FOR STATE-DEPENDENT DELAYS

Alternatively, the following PDE representation can be adopted for (110)–(111):

$$\partial_t u_{av}(x, t) = -\partial_x u_{av}(x, t), \quad x \in [0, D(G_{av}(t))], \quad (156)$$

$$u_{av}(0, t) = U_{av}(t), \quad (157)$$

$$\dot{G}_{av}(t) = H u_{av}(D(G_{av}(t)), t). \quad (158)$$

Introducing the change of variables:

$$z = \frac{D(G_{av}(t)) - x}{D(G_{av}(t))}, \quad (159)$$

the system (156)–(158) can be rewritten as

$$\partial_t v_{av}(z, t) = \lambda_v(z, t) \partial_z v_{av}(z, t), \quad z \in [0, 1], \quad (160)$$

$$v_{av}(1, t) = U_{av}(t), \quad (161)$$

$$\dot{G}_{av}(t) = H v_{av}(0, t), \quad (162)$$

where

$$\lambda_v(z, t) = \frac{1 - z \dot{D}(G_{av}(t))}{D(G_{av}(t))} = \frac{1 - z H \nabla D(G_{av}(t)) v_{av}(0, t)}{D(G_{av}(t))}. \quad (163)$$

As expected, the transport speed is a function of the spatial variable z , the average state G_{av} , and the delayed input $u_{av}(0, t)$. Moreover, the system (160)–(162) is well-posed if the transport speed $\lambda_v(z, t)$ is strictly positive, that is:

$$\lambda_v(z, t) = 1 - z H \nabla D(G_{av}(t)) v_{av}(0, t) > 0, \quad (164)$$

$\forall t > 0, z \in (0, 1), D(\cdot) > 0$. Noting that the function (164) is a linear function of z , its extrema are given by $\lambda_v(0, t)$ and $\lambda_v(1, t)$. Thus,

$$\lambda_v(0, t) = 1, \quad (165)$$

$$\lambda_v(1, t) = 1 - H \nabla D(G_{av}(t)) v_{av}(0, t), \quad (166)$$

leading to the following feasibility condition:

$$F_v(t) = H \nabla D(G_{av}(t)) v_{av}(0, t), \quad (167)$$

$$F_v(t) < \bar{c}, \quad 0 < \bar{c} < 1. \quad (168)$$

Defining the predictor feedback control law of $p(x, t)$ as:

$$U_{av}(t) = k p(D(G_{av}(t)), t), \quad (169)$$

$$p(x, t) = G_{av}(t) + H \int_0^x u_{av}(s, t) ds, \quad x \in [0, D(G_{av}(t))], \quad (170)$$

and applying again the change of variables (159), the following equivalent predictor feedback control law is obtained for the normalized system (160)–(162)

$$U_{av}(t) = k p_v(1, t), \quad (171)$$

$$p_v(z, t) = G_{av}(t) - D(G_{av}(t)) H \int_0^z v_{av}(s, t) ds, \quad z \in [0, 1]. \quad (172)$$

Here, $p(x, t)$ and $p_v(z, t)$ are the predictor states of the systems (156)–(158) and (160)–(162), respectively, and $k > 0$ is the linear feedback gain.

For the stability analysis, let us consider the normalized PDE representation of (110)–(111) given by (160)–(162). We introduce the backstepping transformation [47]

$$w_{av}(z, t) = v_{av}(z, t) - k p_v(z, t), \quad (173)$$

where $p_v(z, t)$ is defined in (172) and its associated inverse transformation is [47]

$$v_{av}(z, t) = w_{av}(z, t) + k p_v(z, t), \quad (174)$$

where

$$\pi_v(z, t) = G_{av}(t) - D(G_{av}(t))H \int_0^z (w_{av}(s, t) + k\pi_v(s, t)) ds, \quad z \in [0, 1]. \quad (175)$$

Applying the backstepping transformation (173) to (160)–(162), the target system is given by

$$\partial_t w_{av}(z, t) = \lambda_w(z, t) \partial_z w_{av}(z, t), \quad z \in [0, 1], \quad (176)$$

$$w_{av}(1, t) = 0 \quad (177)$$

$$\dot{G}_{av}(t) = kHG_{av}(t) + Hw_{av}(0, t), \quad (178)$$

where

$$\lambda_w(z, t) = \frac{1 - zH\nabla D(G_{av}(t))(kG_{av}(t) + w_{av}(0, t))}{D(G_{av}(t))}. \quad (179)$$

Considering the Lyapunov functional

$$L(t) = \frac{1}{2} \int_0^1 e^{gz} w_{av}(z, t)^2 dz, \quad (180)$$

we can calculate its time-derivative as follows

$$\begin{aligned} \dot{L}(t) &= \int_0^1 e^{gz} w_{av}(z, t) \partial_t w_{av}(z, t) dz \\ &= \int_0^1 e^{gz} w_{av}(z, t) \lambda_w(z, t) \partial_z w_{av}(z, t) dz \\ &= \left[e^{gz} \lambda_w(z, t) w_{av}(z, t)^2 \right]_0^1 \\ &\quad - \int_0^1 (g e^{gz} \lambda_w(z, t) + e^{gz} \partial_z \lambda_w(z, t)) w_{av}(z, t)^2 dz \\ &= e^g \lambda_w(1, t) w_{av}(1, t)^2 - \lambda_w(0, t) w_{av}(0, t)^2 \\ &\quad - \int_0^1 e^{gz} (g \lambda_w(z, t) + \partial_z \lambda_w(z, t)) w_{av}(z, t)^2 dz. \end{aligned} \quad (181)$$

By definition

$$\partial_z \lambda_w(z, t) = \frac{-\dot{D}(G_{av}(t))}{D(G_{av}(t))} \quad (182)$$

and

$$\begin{aligned} \Phi(z) &= g \lambda_w(z, t) + \partial_z \lambda_w(z, t) \\ &= \frac{g - (1 + gz) \dot{D}(G_{av}(t))}{D(G_{av}(t))}. \end{aligned} \quad (183)$$

Accounting for the derivative of the state-dependent delay, we can rewrite (183) into:

$$\Phi(z) = \frac{g - (1 + gz)F_w(t)}{D(G_{av}(t))}, \quad (184)$$

where the feasibility condition is ensured if:

$$F_w(t) = H\nabla D(G_{av}(t))(kG_{av}(t) + w_{av}(0, t)), \quad (185)$$

$$F_w(t) < \bar{c}, \quad 0 < \bar{c} < 1. \quad (186)$$

Noting that the function (184) is a linear function of z , its minimum is given by $\Phi(0)$ or $\Phi(1)$. Thus,

$$\Phi(x) \geq \min\{\Phi(0), \Phi(1)\}, \quad (187)$$

$$\min\{\Phi(0), \Phi(1)\} = \frac{\min\{g + F_w(t), (1 + g)F_w(t)\}}{D(G_{av}(t))}. \quad (188)$$

Recalling the prediction time $\sigma(t) = t + D(G_{av}(t))$ is always greater than the current time t , we can also show

$$\Phi(0) \geq \frac{g + \bar{c}}{D(G_{av}(t))}, \quad (189)$$

$$\Phi(1) \geq \frac{(1 + g)\bar{c}}{D(G_{av}(t))}, \quad \forall g > 0. \quad (190)$$

To preserve the causality of the system, the state-dependent delay function $D \in C^1(\mathbb{R}; \mathbb{R}_+)$ must be positive and therefore the function $\frac{1}{D(G_{av}(t))}$ for all $G_{av} \in \mathbb{R}$ admits a positive minimum g_0^* which implies that

$$g_0^* \Phi^* \leq \min\{\Phi(0), \Phi(1)\} \quad (191)$$

$$\Phi^* = \min\{(g + \bar{c}), (1 + g)\bar{c}\} = (1 + g)\bar{c}, \quad \forall \bar{c} < 1. \quad (192)$$

Choosing $g > 0$, from (177) and (181), the following inequality can be established

$$\dot{L}(t) \leq -g_0^* w_{av}(0, t)^2 - \int_0^1 g_0^* \Phi^* e^{gx} w_{av}(x, t)^2 dx < 0. \quad (193)$$

Hence, the exponential stability of the complete average closed-loop system (160)–(162) and (171)–(172) can be guaranteed with the following Lyapunov-Krasovskii functional

$$V(t) = \frac{G_{av}^2(t)}{2} + L(t) + \frac{1}{2} w_{av}^2(1, t), \quad (194)$$

reminding that $G_{av}(t) = H\tilde{\theta}_{av}(t)$ from (108). It means that $G_{av}(t) \rightarrow 0$ implies $\tilde{\theta}_{av}(t) \rightarrow 0$, as desired.

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GEORGE CARNEIRO SANTOS was born in Rio de Janeiro, Brazil, in 1989. He received the bachelor's degree in control and automation engineering from the Federal Center for Technological Education Celso Suckow da Fonseca (CEFET-RJ), Rio de Janeiro, in 2013, and the M.Sc. degree in electronics engineering from the State University of Rio de Janeiro (UERJ), Rio de Janeiro, in 2017. He is currently an Access Management with the Globenet Cabos Submarinos S. A., Rio de Janeiro.

His current research interests include nonlinear systems, time delays, adaptive systems, extremum seeking, automation, and automatic control.



TIAGO ROUX OLIVEIRA (Senior Member, IEEE) was born in Rio de Janeiro, Brazil, in 1981. He received the B.Sc. degree in electrical engineering from the State University of Rio de Janeiro (UERJ), in 2004, and the M.Sc. and Ph.D. degrees in electrical engineering from the Graduate School and Research in Engineering, Federal University of Rio de Janeiro (COPPE/UFRJ), in 2006 and 2010, respectively.

In 2014, he was a Visiting Scholar with the University of California, San Diego (UCSD), San Diego, CA, USA. He is currently an Associate Professor with the Department of Electronics and Telecommunication Engineering and a Researcher with the Graduate Program of Electronics Engineering (PEL), UERJ. He published more than 50-refereed Journal articles and more than 100-refereed conference papers. His current research interests include nonlinear control theory, time-delay systems, partial differential equations, adaptive control systems, extremum seeking, sliding mode control, state estimation and output feedback, robotics, and real-time control systems.

Dr. Oliveira was a recipient of the Bolsa Nota 10 sponsored by the Brazilian Agency FAPERJ, the CAPES National Award of Best Thesis in

electrical engineering, in 2011, and the FAPERJ Young Researcher Award, in 2012, 2015, and 2018. He has served as a member for the International Federation of Automatic Control (IFAC) Technical Committees: Adaptive and Learning Systems (TC 1.2) and Control Design (TC 2.1), and the Technical Committee on Variable Structure and Sliding Mode Control of the IEEE Control Systems Society (CSS). In 2017, he was nominated as an Affiliate Member of the Brazilian Academy of Sciences (ABC). In 2018, he was elevated to the grade of the IEEE Senior Member of the CSS. In 2020, he was elected and nominated as the Chair of the Technical Committee TC 1.2 (Adaptive and Learning Systems) of the IFAC for the triennial 2020-2023. He was the Guest Editor of the *International Journal of Adaptive Control and Signal Processing* in the Special Issue From Adaptive Control to Variable Structure Systems—Seeking Harmony and one of the organizers of the session Smart Cyber-Physical Systems for Restoring Human Movement After Paralysis of the 1st IFAC Conference on Cyber-Physical and Human-Systems (2016). He has served also as an Associate Editor for the *Journal of the Franklin Institute*, the *Journal of Control, Automation, and Electrical Systems*, the IEEE LATIN AMERICA TRANSACTIONS, and *Systems & Control Letters*. He has served as an Advisor for the IEEE Robotics and Automation Society Student Branch Chapter.

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